MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

LOGIC UNIT, ARITHMETIC, DIGITAL, KEY CODE RDH

This amendment forms a part of Military Specification MIL-M-28787/79A(EC), dated 31 July 1979, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 3, 4, AND 5

1.2, Propagation delay time (t_{PLH}) and (t_{PHL}): Delete in their entirety.

PAGE 6

1.3d: Delete in its entirety and substitute:

"d. The module also includes a look-ahead carry generator (LCG) capable of generating a carry across four binary adders or groups of adders. The LCG can be cascaded to perform full look-ahead across N-bit adders. Positive logic equations for the LCG are:

(1) $C_{N+X} = G0+P0C_N$.

(2) $C_{N+Y} = G1+P1G0+P1P0C_N$.

(3) $C_{N+Z} = G2+P2G1+P2P1G0+P2P1P0C_N$.

(4) $\overline{G} = \overline{G3+P3G2+P3P2G1+P3P2P1G0}$.

(5) $\overline{P} = \overline{P3P2P1P0}$."

PAGE 23

* 4.3, delete and substitute the following:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

PAGE 32

* FIGURE 7: Delete and substitute new figure 7 as printed on page 3 of this amendment.

PAGE 35

FIGURE 8, Item PO1-2 box: Delete "5.5 V ±1%" and substitute "4.0 V ±1%.

PAGE 43

* TABLE X, note 2, second line: Delete "voltmeter (V03)" and substitute "item M02".

PAGE 44

* FIGURE 11: Delete and substitute new figure 11 as printed on page 4 of this amendment.

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PAGE 47

* FIGURE 12: Delete and substitute new figure 12 as printed on page 5 of this amendment.

PAGE 50

* FIGURE 13: Delete and substitute new figure 13 as printed on page 6 of this amendment.

PAGE 115

FIGURE 25: Delete and substitute new figure 25 as printed on page 7 of this amendment.

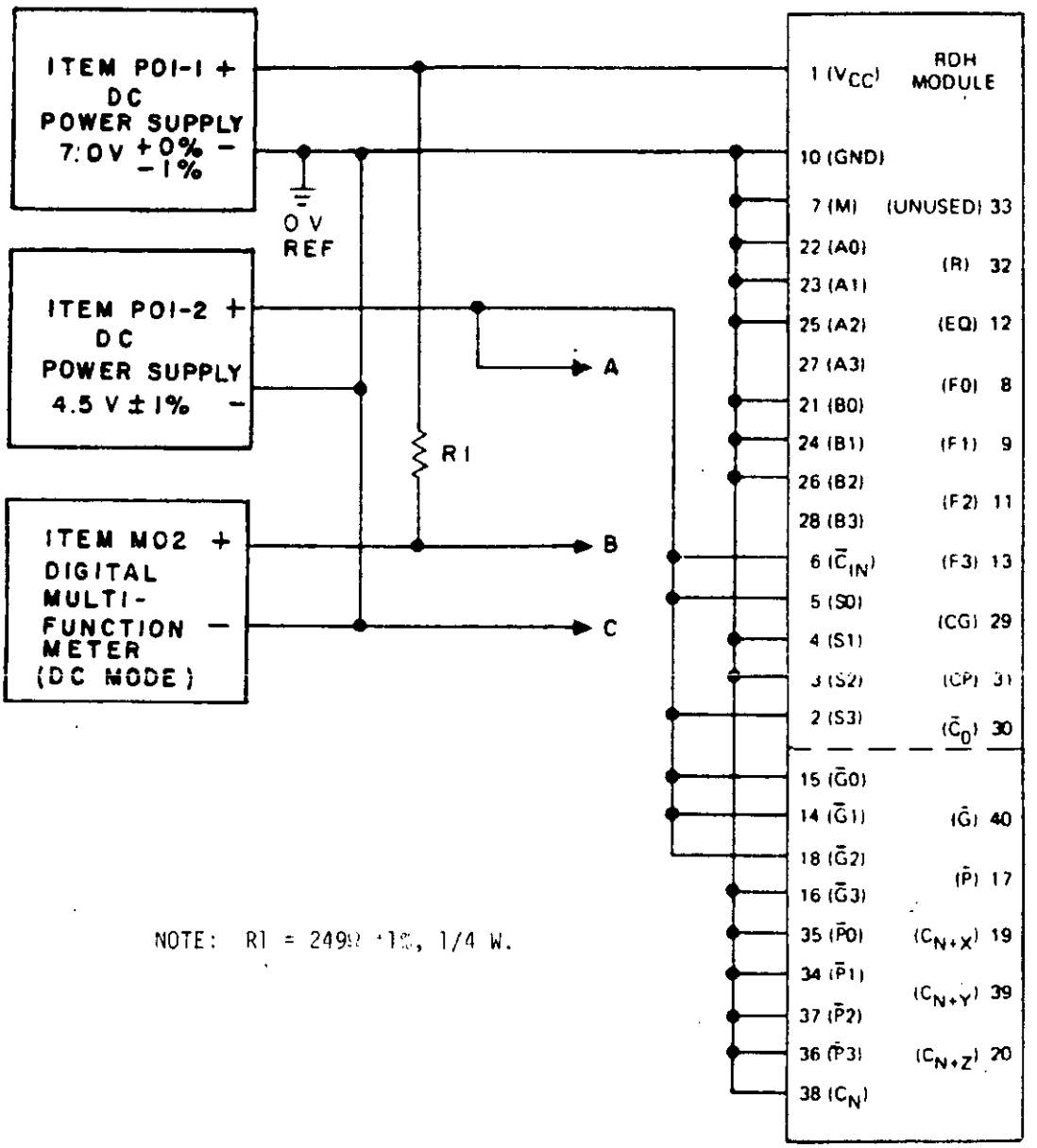
NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N576)

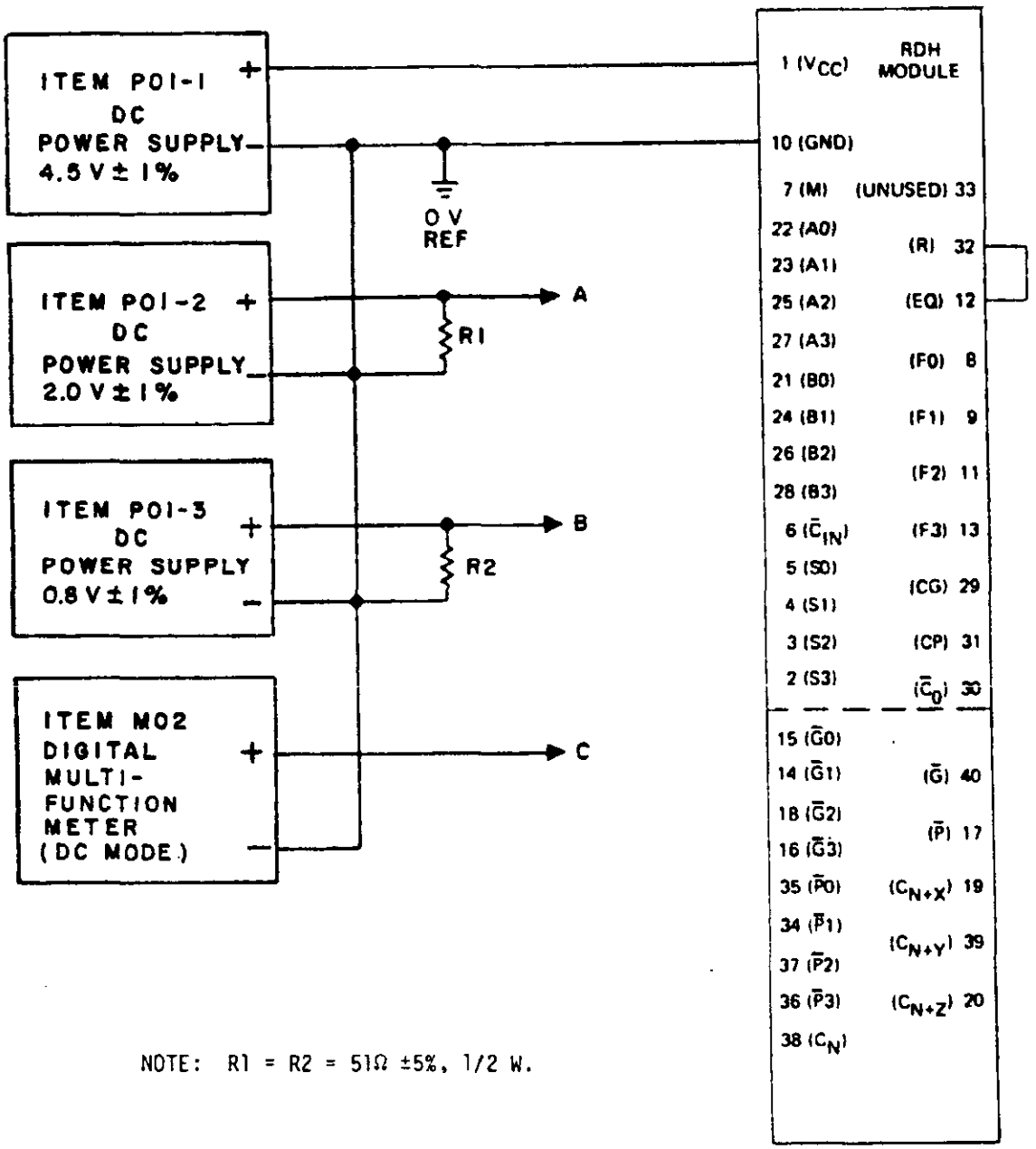
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NOTE: R1 = 249Ω ±1%, 1/4 W.

FIGURE 7. V_{OUT} (low-loaded) test circuit for table VI.

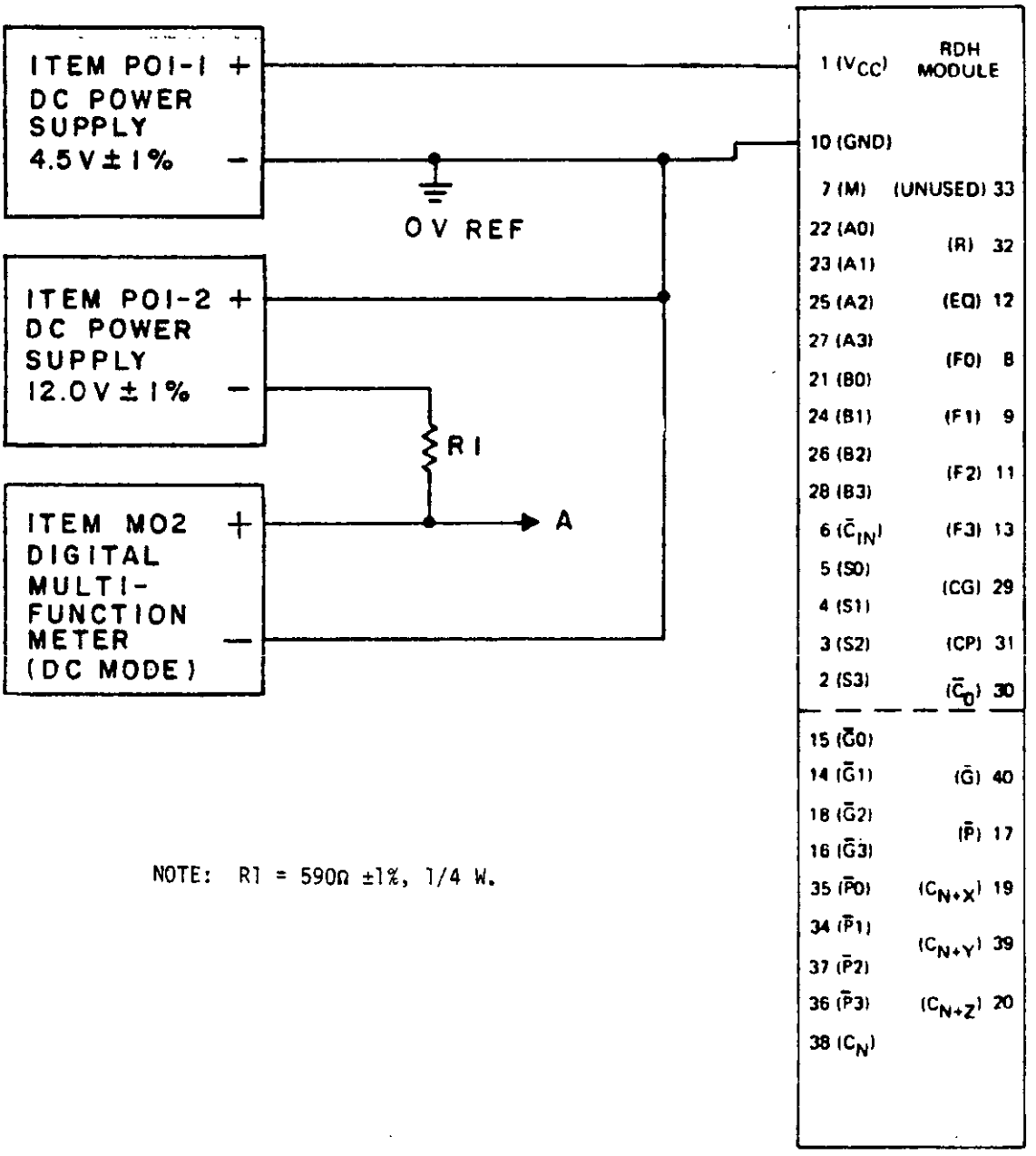
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 11. Functional test circuit for table X.

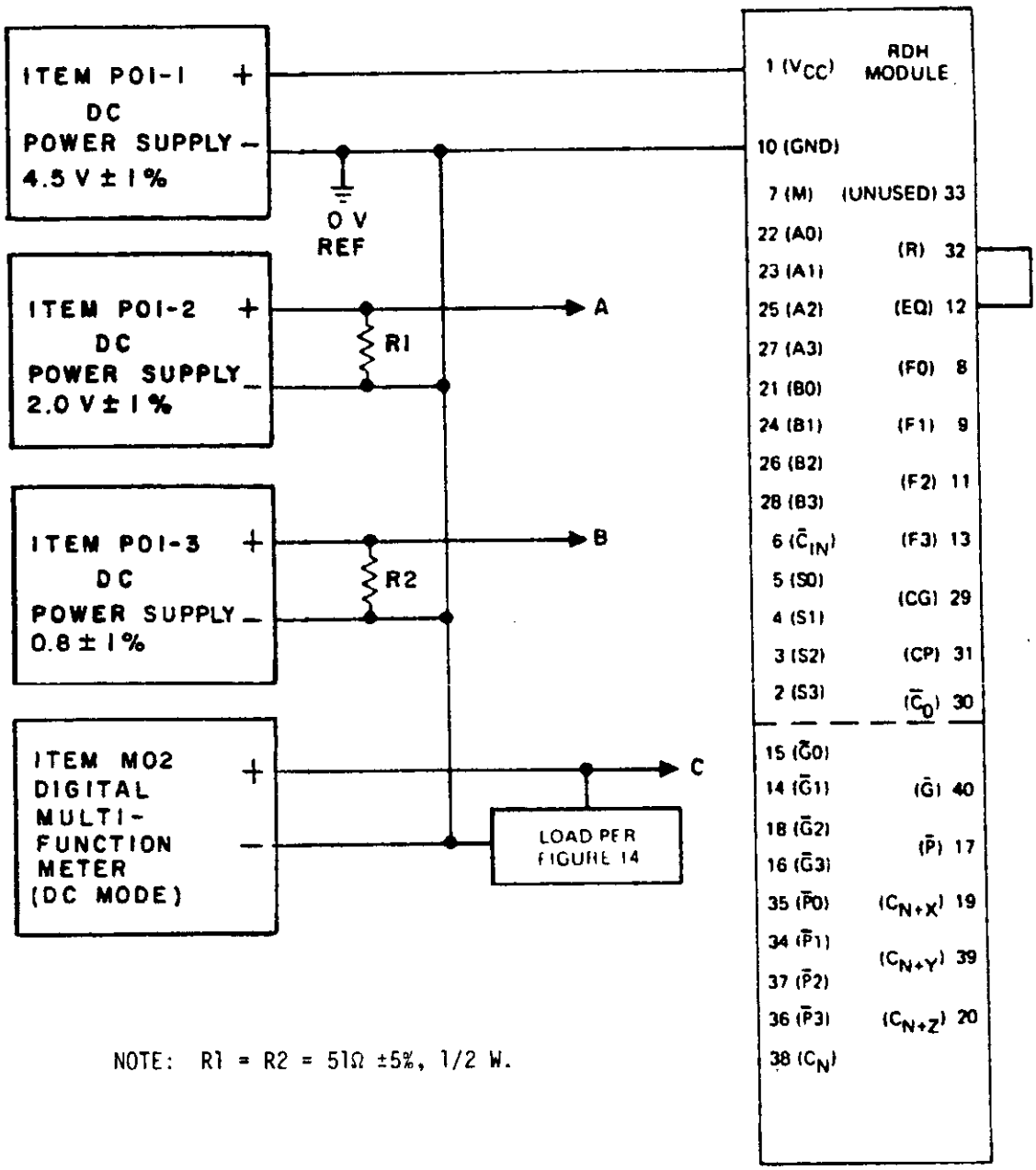
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NOTE: R1 = 590Ω ±1%, 1/4 W.

FIGURE 12. Input clamp diode voltage test circuit for table XI.

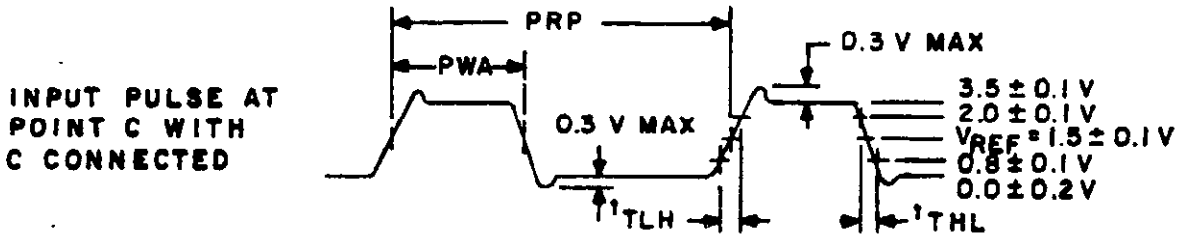
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 13. Static output voltage (low-level and high-level) test circuit for tables XII and XIII.

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E IS THE MAXIMUM AMPLITUDE OF THE MOST POSITIVE PEAK MEASURED AFTER A SETTLE TIME OF 45 ns HAS ELAPSED AFTER ANY POSITIVE OR NEGATIVE TRANSITION OF THE INPUT PULSE



NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
   PWA = 250 ±25 ns.
   PRP = 500 ±50 ns.
2. The lead length of electrical conductors between resistor R1 and input pins; between probes YO1-1 and YO1-2 and input and output pins; between load circuits and output pins; and between ground leads of probes YO1-1 and YO1-2 and test fixture and module ground pins, shall be less than 1 inch (25.4 mm).

FIGURE 25. Output crosstalk duration test circuit waveforms for table XIX.

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